

Introduction

Set: (language) : It is a collection of objects.

unordered

(well defined)

members

cardinality

(size of set $\rightarrow$ finite)

distinct

Domain

Set types:

1. Finite and Infinite set.
2. Empty set and Universal set.
3. Countable and Uncountable set.
- 4.

Set Operations:

- Union
- Intersection
- Complement
- Set difference

$$A \cup B = \{x | x \in A \text{ or } x \in B\} \quad \text{let } |A|=m, |B|=n, \max(m,n) \leq |A \cup B| \leq m+n$$

$$A \cap B = \{x | x \in A, x \in B\} \quad 0 \leq |A \cap B| \leq \min(m,n)$$

$$\bar{A} = \{x | x \notin A\} = \{x | x \in U, x \notin A\}$$

$$A - B = \{x | x \in A, x \notin B\} = A \cap \bar{B}$$

Basics of TOC:

Type 0 - (formal languages)

Type 1 - formal languages

Type 2 - formal languages

Type 3
Regular
languages
(FA)

CFLs, PDA, CFGs

CSLs, LBA, CSGs

Recursively Enumerable sets, unrestricted
(Turing machine) grammars

• Set of all regular languages

$\subset$

Set of all context free languages

$\subset$

Set of all context sensitive language

$\subset$

Set of all Recursively Enumerable languages.

• Type 3 $\subset$ Type 2 $\subset$ Type 1 $\subset$ Type 0

• FA $\subset$ PDA $\subset$ LBA $\subset$ TM

• $LS(FA) \subset LS(PDA) \subset LS(LBA) \subset LS(TM)$

Symbols: Raw thing that can not be broken into any other symbol. (It is one length)

Alphabets (Σ): Set of (finite no. of) symbols.

Strings: Sequence of (finite) symbols called as "string"

$\Sigma^0 = \{\epsilon\}$ = set of all zero length string.

Σ^1 = set of all one length strings.

Σ^2 = set of all two length strings.

$\Sigma = \{a, b\}$

↓ $|w|=0$ ↓ $|w|=1$ ↓ $|w|=2$ ↓ $|w|=3$ ↓
 $\lambda \in \epsilon$ a aa aaa
 b ab bba bbb
 ba bbb
 bb bbb

Language: Set of strings. $|\{\epsilon\}| = |\{\epsilon\}| = 0$ $|\{ \}$ = 0
 set ↗ string size of set

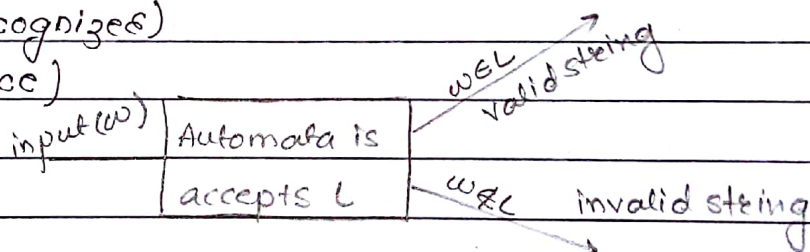
Formal language: It is a language where form of string is known.

Four types of languages have logic to compute.

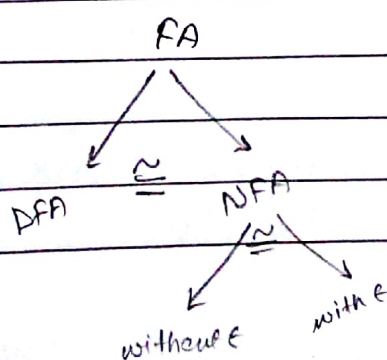
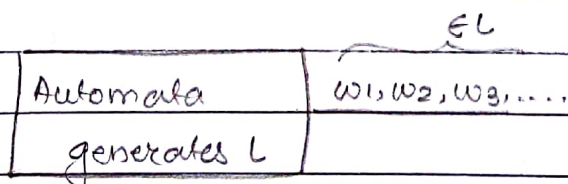
Automata:

1. Acceptor (Recognizes)

(Universal acceptance)



2. produces / Generator / Enumerator / Transducer



PDA
 ↓ called as
 NPDA
 $DPDA \subset PDA (NPDA)$
 $L(DPDA) \subset L(PDA)$
 Set of DCFLs $\subset$ Set of CFLs

